

Quantum Computing

Lecture $|01\rangle$

A review of complex linear algebra

Paolo Zuliani



Outline

- Complex numbers
- Vectors and matrices (of complex numbers)
- Eigenvectors and eigenvalues

Why Complex Numbers

- The set of the **natural numbers**

$$\mathbb{N} = \{1, 2, 3, \dots\}$$

is the most fundamental object in mathematics.

- Note that the simple equation

$$x + 21 = 7$$

has no solution in the naturals!

- However, if we allow '**negative**' numbers we can solve and obtain $x = 7 - 21 = -14$



Why Complex Numbers

- Therefore, we get the set of the **integers**

$$\mathbb{Z} = \{ \dots, -3, -2, -1, 0, 1, 2, \dots \}$$

- Note:
 - $x + a = b$ can now be solved for *any* integers a, b
 - The solution $x = b - a$ is always an integer!

Why Complex Numbers

- Hold on! How about solving

$$22x + 7 = 0$$

Does it have integer solutions?

- In general, no. We must introduce **rational** numbers

$$x = \frac{-7}{22}$$



Why Complex Numbers

- Therefore, we get the set of the **rational**s

$$\mathbb{Q} = \{ \dots, -\frac{2}{3}, -2, \dots, -\frac{1}{3}, -\frac{1}{2}, -1, 0, 1, \frac{1}{2}, \frac{1}{3}, \dots, 2, \frac{2}{3}, \dots \}$$

- Note:
 - $ax + b = 0$ can now be solved for *any* rationals a, b (except $a = 0$)
 - The solution $x = -\frac{b}{a}$ is always a rational!

Why Complex Numbers

- Now consider:

$$ax^2 + bx + c = 0 \text{ where } a \neq 0, b, c \text{ are integers}$$

- The solutions are:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \quad \text{as long as } \Delta = (b^2 - 4ac) \geq 0$$

- If Δ is the square of an integer, then solutions are rational
- However, solutions may be **irrational** (e.g., $\Delta = 2$)



Why Complex Numbers

- How about when $(b^2 - 4ac) < 0$?? (A legitimate question!)
- Square root of a negative number?!

$$\sqrt{-4} = \sqrt{4}\sqrt{-1} = 2\sqrt{-1} = 2i$$

where $i = \sqrt{-1}$ is an **imaginary number** (note $i^2 = -1$)

- **Complex number:** $a + ib$ where a, b are reals
 - Example: $3.17 - i5$ is a complex number

Do We Need More Numbers?

- How about solving

$$4x^5 + x^4 + 3.44x^2 + x - 7 = 0$$

- Thankfully, complex numbers are enough!
- **[Fundamental Theorem of Algebra]:**

Every equation

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 = 0$$

with $a_n \neq 0$ and all a_i 's complex has **exactly** n complex solutions



Complex Numbers

- The set of complex numbers is denoted by $\mathbb{C}$

$$\mathbb{C} = \{a + ib, \text{ where } a, b \text{ are real}\}$$

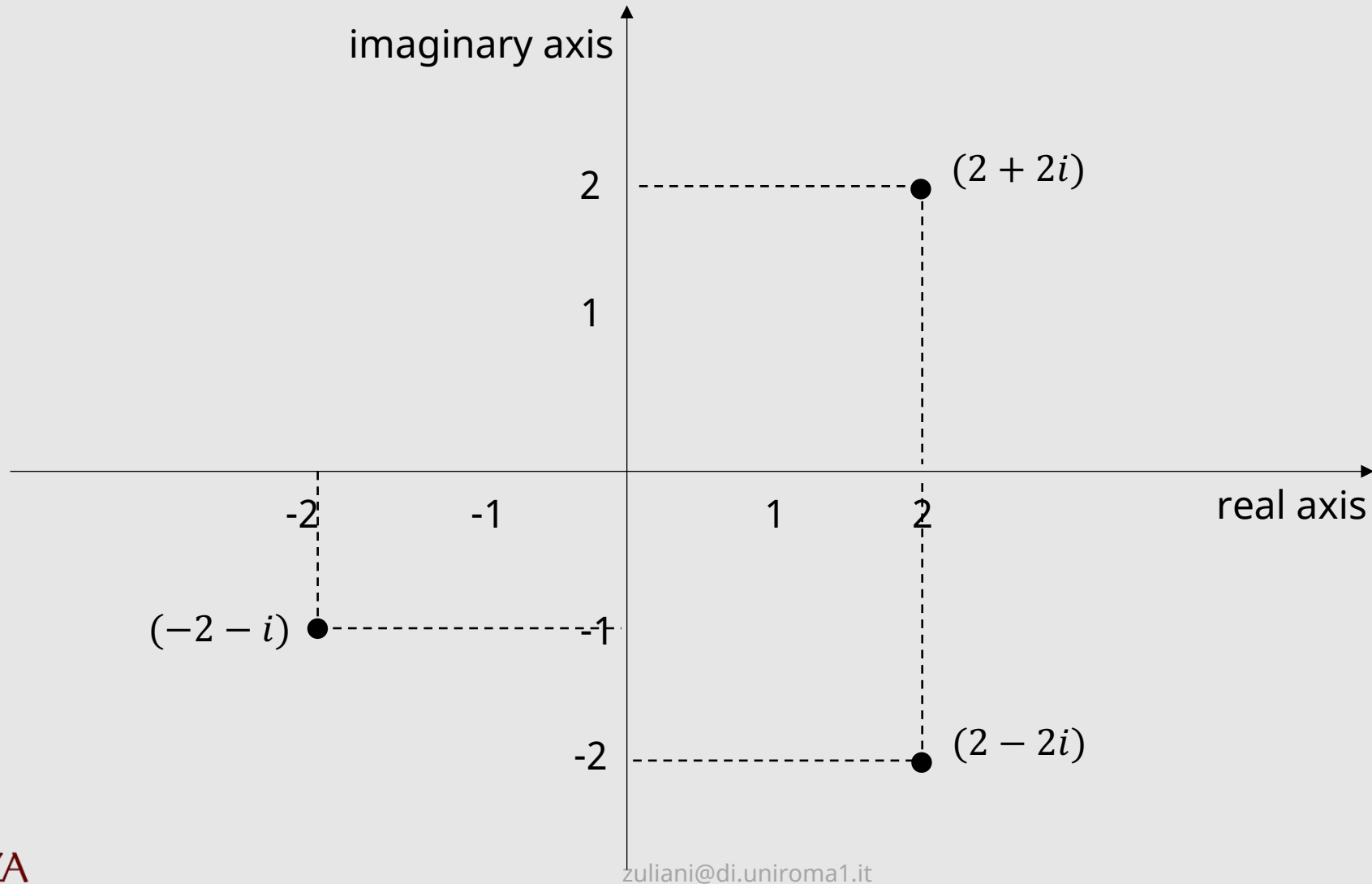
- Given a complex $z = a + ib$ we define

$$\operatorname{Re}(z) = a \quad (\text{the } \mathbf{real} \text{ part of } z)$$

$$\operatorname{Im}(z) = b \quad (\text{the } \mathbf{imaginary} \text{ part of } z)$$

Note: the imaginary part of a complex number is a real number!

Complex Numbers: How to Plot Them



Complex Numbers: Basic Operations

$$(a + ib) + (c + id) = (a + c) + i(b + d)$$

$$(a + ib) * (c + id) = (ac - bd) + i(ad + bc)$$

The **conjugate** of $z = (a + ib)$ is $\bar{z} = \overline{(a + ib)} = (a - ib)$

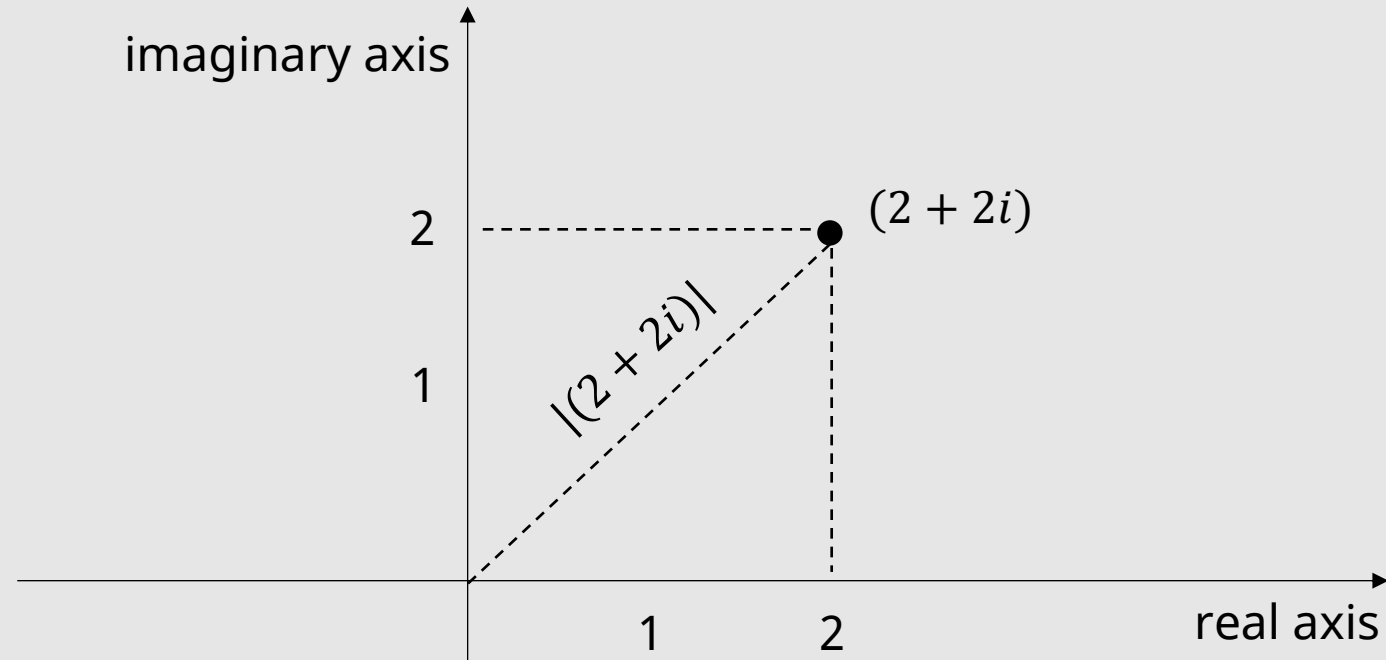
The **modulus** of $z = (a + ib)$ is $|z| = \sqrt{a^2 + b^2}$

- $|z|$ is real!

Note that $|zw| = |z||w|$ and $|z| = \sqrt{z\bar{z}}$ (exercise)



Complex Numbers: Basic Operations



- How about complex division $\frac{z}{w}$?
- Multiply both numerator and denominator by $\bar{w}$...

Complex Numbers: Polar Coordinates

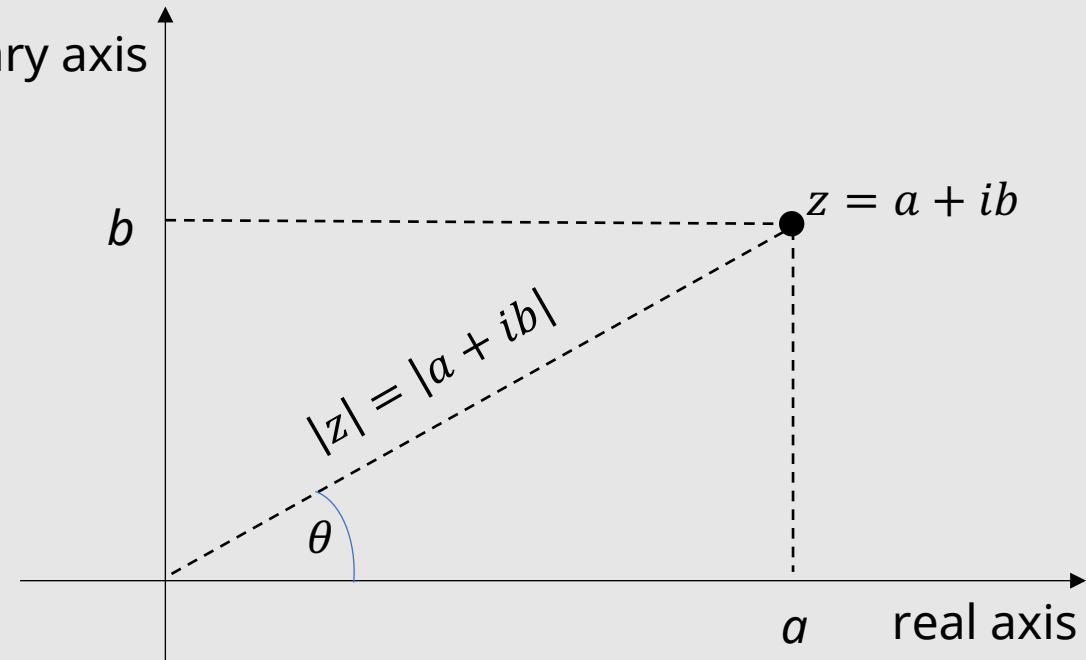
- Suppose $z = a + ib$ is non-zero, and so $|z| = \sqrt{a^2 + b^2}$

- Then there is a unique angle θ in the interval $(-\pi, \pi]$ such that:

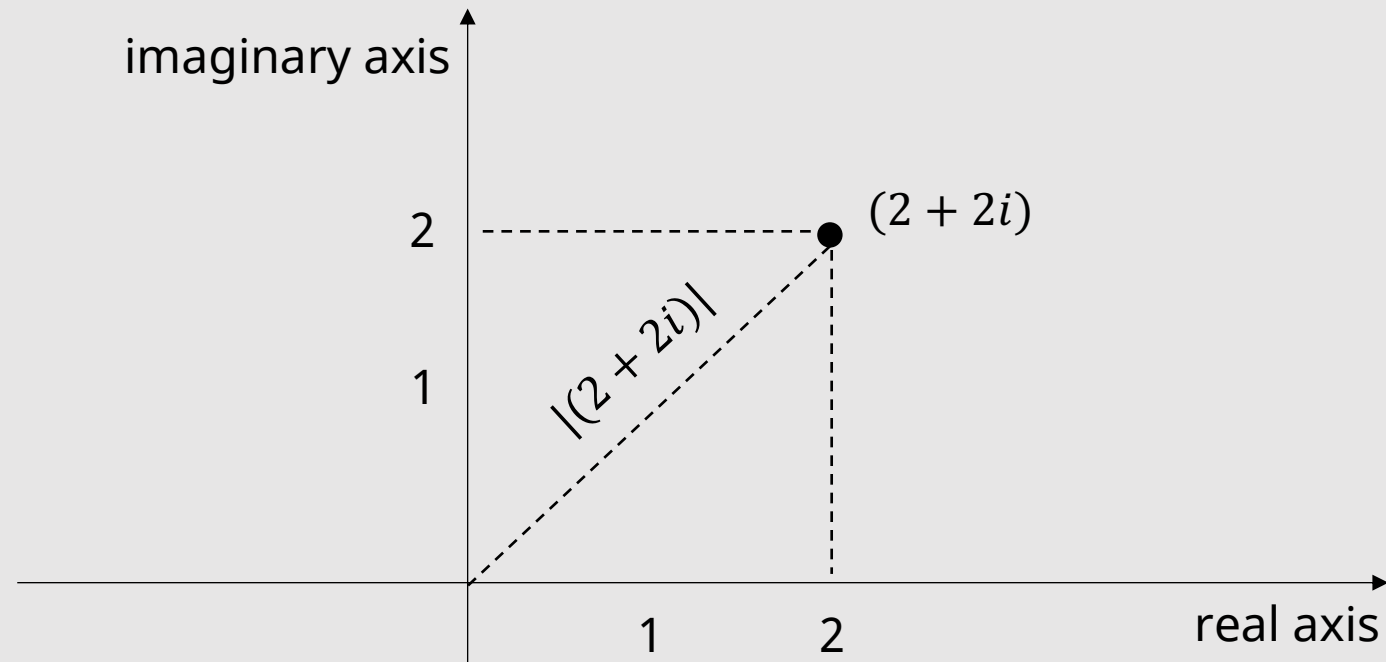
$$\cos \theta = \frac{a}{\sqrt{a^2 + b^2}} = \frac{a}{|z|} \quad \sin \theta = \frac{b}{\sqrt{a^2 + b^2}} = \frac{b}{|z|}$$

- Thus, we can write

$$z = |z|(\cos \theta + i \sin \theta)$$



Complex Numbers: Polar Coordinates



$$\text{For example: } (2 + 2i) = |2 + 2i| \left(\cos \frac{\pi}{4} + i \sin \frac{\pi}{4} \right) = 2\sqrt{2} \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right)$$

Complex Numbers: Polar Coordinates

- We also define the exponential of an imaginary number

$$e^{i\theta} = \cos \theta + i \sin \theta$$

and we can thus write

$$z = |z|e^{i\theta}$$

- This form is handy when dealing with powers of complex numbers:

$$z^2 = (|z|e^{i\theta})^2 = |z|^2 e^{i2\theta}$$

$$\frac{1}{z} = z^{-1} = (|z|e^{i\theta})^{-1} = |z|^{-1} e^{-i\theta}$$

$$\sqrt{z} = (|z|e^{i\theta})^{\frac{1}{2}} = \sqrt{|z|} e^{i\frac{\theta}{2}} \quad (\text{square root always exists!})$$

Vectors

- Simply, lists of (complex) numbers

$$v = \begin{pmatrix} 4 \\ i \\ 2 + 3i \end{pmatrix} \quad w = \begin{pmatrix} 0 \\ 1 \\ i \end{pmatrix}$$

$$v + w = \begin{pmatrix} 4 \\ i \\ 2 + 3i \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ i \end{pmatrix} = \begin{pmatrix} 4 \\ 1 + i \\ 2 + 4i \end{pmatrix}$$

summing vectors

$$v * 2i = 2i * \begin{pmatrix} 4 \\ i \\ 2 + 3i \end{pmatrix} = \begin{pmatrix} 8i \\ -2 \\ -6 + 4i \end{pmatrix}$$

scalar multiplication

Vectors

- The **norm** of a vector is its 'size'
- Given a complex vector of dimension n :

$$v = (v_1, v_2, \dots, v_n)$$

The norm of v is $\|v\|$

$$\|v\| = \sqrt{|v_1|^2 + |v_2|^2 + \dots + |v_n|^2} = \sqrt{\sum_{i=1}^n |v_i|^2}$$

$$\text{Example: } \|(4, i, 2 + 3i)\| = \sqrt{16 + 1 + 13} = \sqrt{30}$$

Matrices

- Essentially: tables of complex numbers, with a 'funky' multiplication rule
- A matrix has **m rows, n columns** (its dimensions are m x n)

- A 3 x 2 matrix:
$$\begin{bmatrix} 3 & i \\ 1 + i & 0 \\ -4 & 2 + 3i \end{bmatrix}$$

- A 2 x 2 matrix (when $m=n$ we say the matrix is square)

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2 x 2 identity matrix

Matrices: Sum

- Two matrices can be *summed* only if they have the **same dimension**; the sum is element-wise (as for vectors)

$$\begin{bmatrix} 3 & i \\ 1+i & 0 \\ -4 & 2+3i \end{bmatrix} + \begin{bmatrix} 1 & i \\ i & 10 \\ -4 & 2-3i \end{bmatrix} = \begin{bmatrix} 4 & 2i \\ 1+2i & 10 \\ -8 & 4 \end{bmatrix}$$

- For any two matrices A, B with the same dimension

$$A + B = B + A \quad (\text{matrix sum is } \mathbf{commutative})$$



Matrices: Row-by-Column Product

- Two matrices A, B

$$\dim(A)=m \times n \quad \dim(B)=p \times q$$

can be *multiplied* only if **$n = p$**

the product $A \cdot B$ has dimension **$m \times q$**

$$\begin{bmatrix} \boxed{3} & \boxed{i} \\ 1+i & 0 \\ -4 & 2+3i \end{bmatrix} \cdot \begin{bmatrix} \boxed{3} & \boxed{1} \\ \boxed{0} & \boxed{0} \end{bmatrix} = \begin{bmatrix} \boxed{9+i0} & 3+i0 \\ 3+3i+0 & 1+i+0 \\ -12+0 & -4 \end{bmatrix} = \begin{bmatrix} 9 & 3 \\ 3+3i & 1+i \\ -12 & -4 \end{bmatrix}$$

$\boxed{3 \times 2} \quad \boxed{2 \times 2} \quad \boxed{3 \times 2}$

Matrices: Row-by-Column Product

$$\begin{bmatrix} 3 & i \\ 1+i & 0 \\ -4 & 2+3i \end{bmatrix} \cdot \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} 3i+i \\ i+i^2 \\ -4i+2+3i \end{bmatrix} = \begin{bmatrix} 4i \\ -1+i \\ 2-i \end{bmatrix}$$

3×2 2×1 3×1

Scalar multiplication is **element-wise**:

$$\begin{bmatrix} 3 & i \\ 1+i & 0 \\ -4 & 2+3i \end{bmatrix} \cdot 2 = \begin{bmatrix} 6 & 2i \\ 2+2i & 0 \\ -8 & 4+6i \end{bmatrix}$$

Matrices: Row-by-Column Product

In general, matrix multiplication is **NOT** commutative:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

but

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$$

This property is **crucially important** in quantum physics (and thus quantum computing!)



Matrices: Eigenvectors & Eigenvalues

A (non-zero) vector v is an **eigenvector** of matrix A with **eigenvalue** λ if

$$A \cdot v = \lambda \cdot v$$

Equivalently: $A \cdot v - \lambda \cdot v = 0$ iff $(A - \lambda I) \cdot v = 0$

Now, $(A - \lambda I) \cdot v = 0$ has non-zero v solutions iff $\det(A - \lambda I) = 0$

Matrices: Eigenvectors & Eigenvalues

Example: $\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ has eigenvalues +1 and -1 with eigenvectors $\begin{bmatrix} 1 \\ i \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -i \end{bmatrix}$, respectively. In fact, for eigenvalue +1:

$$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} -i^2 \\ i \end{bmatrix} = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

Exercise: prove the above.

Matrices: Eigenvectors & Eigenvalues

Note that:

- in general, both v and λ are complex;
- it is possible for an eigenvalue to be associated to **more than one** eigenvector [such eigenvectors form a linear subspace – more later]

Spoiler: quantum mechanics models physical measurements with certain linear operators (matrices) that always have **real** eigenvalues

